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Agentic AI for Information Retrieval

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Abstract

This document presents the homework of the *Information Retrieval* course for the academic year 2024/2025, concerning the use of AI agents for Information retrieval tasks.

1 Problem Introduction

Agentic AI refers to a class of artificial intelligence systems designed to operate with autonomy, pursue complex goals, and adaptively interact with dynamic environments. Unlike traditional AI models that passively respond to input, agentic systems proactively seek information, decompose tasks, devise strategies, and execute actions—often through the integration of tools, APIs, and external data sources. These systems ingest and analyze vast amounts of data, enabling them to make decisions with limited supervision and act independently across diverse contexts, including robotics, software development, and virtual assistance[1]. Agentic AI combines the generative and flexible capabilities of large language models (LLMs) with the structured accuracy of traditional programming paradigms. While generative AI is primarily reactive—producing content such as text, images, or code in response to prompts—agentic AI exhibits agency, meaning it can autonomously plan and execute actions toward a goal using techniques such as natural language processing, machine learning, reinforcement learning, and knowledge representation[2]. In the domain of Information Retrieval (IR), the integration of agentic capabilities represents a paradigm shift from static document ranking to interactive, context-aware, and user-aligned retrieval systems. Agentic IR agents can dynamically refine queries, traverse knowledge graphs, interact with databases, validate factual information, and even invoke external APIs to augment results. For instance, such a system might autonomously break down a complex information need into sub-questions, retrieve relevant documents from distributed sources, and synthesize a coherent answer. These capabilities are actively being explored in current AI research and industry initiatives, such as OpenAI's AutoGPT and Microsoft's Jarvis, which aim to extend the reasoning and planning abilities of LLMs into more autonomous, decision-making agents.

[1] NVIDIA Agentic AI
[2] IBM Agentic AI

2 State-of-the-Art

The integration of agentic AI into information retrieval (IR) systems has catalyzed the development of autonomous agents capable of dynamic reasoning, tool utilization, and adaptive interaction with complex data environments.

Agentic Information Retrieval (Agentic IR): This conceptual framework reimagines IR by leveraging large language models (LLMs) as autonomous agents. These agents are designed to perform multi-step reasoning, employ retrieval-augmented generation (RAG), and utilize tools such as web search and code execution to fulfill complex information needs. The approach emphasizes techniques like prompt engineering, reflection, and multi-agent collaboration to enhance retrieval performance and adaptability [1].

Agentic Reasoning Framework: This framework enhances LLM reasoning by integrating external tool-using agents. It introduces a "Mind Map" agent that constructs structured knowledge graphs to track logical relationships, improving deductive reasoning. By incorporating web search and coding agents, the system enables real-time retrieval and computational analysis, demonstrating significant improvements in complex scientific reasoning tasks [2].

PaperQA performs information retrieval across full-text scientific articles, assesses the relevance of sources, and synthesizes answers. It surpasses existing LLMs and agents on science QA benchmarks and introduces LitQA, a benchmark requiring retrieval and synthesis from full-text scientific papers [3].

CollEX: This multimodal agentic RAG system facilitates interactive exploration of extensive scientific collections. By employing large vision-language models as multimodal agents within a chat interface, CollEX supports curiosity-driven exploration, integrating textual and visual modalities to aid educators and researchers in accessing diverse scientific records [4].

Agentic Contextual Retrieval in Telecommunications: In the realm of intelligent communications and networking, agentic AI enables AI-driven agents to perceive, reason, decide, and act within dynamic networking environments. By integrating multi-source retrieval, structured reasoning, and self-reflective validation, agentic contextual retrieval frameworks enhance decision-making in telecom applications, such as network planning and resource allocation [5].

These advancements illustrate the transformative potential of agentic AI in information retrieval, enabling systems to move beyond static query-response models toward dynamic, context-aware, and goal-directed information access.

3 Project Information

This project aims to develop a strategy and framework in the domain of agentic artificial intelligence for information retrieval systems. The system will leverage agentic capabilities to perform one or more of the tasks described in Table 1.

3.1 Data

A comprehensive collection of open-source datasets for training and evaluating agentic AI models is available at this site. Some of the most widely used datasets, along with a brief overview of their characteristics, are summarized in Table 2.

4 Project Document Structure

Students must provide all the assignment materials to the teacher **at most 7 days before the exam date** they want to participate to. Project files to deliver must include:

- Source code of the project (in a compressed file or a Github repository)
- Project Report as PDF file
- Configuration files and additional documentation, if any.

The project document follows a structured approach to guide the development of an Agentic AI system for Information Retrieval. The content will be adapted based on the specific retrieval task chosen by the student while ensuring a comprehensive discussion of data, methodologies, system design, and

Task Description	Goal in Information Retrieval
Autonomous Query Refinement	
An agentic IR system generates and iteratively refines search queries by analyzing user intent, intermediate results, and feedback loops. It may decompose broad information needs into sub-queries, rephrase terms, or expand synonyms to narrow or broaden the search.	Improve retrieval precision and recall by dynamically adapting queries to user goals and contextual signals, reducing reliance on manual query engineering.
Multi-Step Retrieval Planning	
This task involves planning a sequence of retrieval actions—such as initial document search, knowledge graph navigation, targeted API calls, and focused subsearches—to address complex information needs. The agent constructs an action plan and adjusts it as new information arrives.	Enable end-to-end, goal-driven retrieval workflows that can handle compound questions, multi-hop reasoning, and cross-source synthesis.
Tool-Augmented Retrieval	
Agents leverage external tools (e.g., web search APIs, code execution engines, databases) within the retrieval loop. They decide which tools to invoke, format inputs, and integrate outputs into their reasoning chain.	Extend the scope of retrieval beyond static corpora by incorporating real-time data, computations, and domain-specific services to enrich answer generation.
Knowledge Graph Traversal	
This task focuses on navigating structured knowledge representations (ontologies, semantic graphs) to retrieve entities, relationships, and context. The agent selects traversal paths, applies reasoning rules, and extracts subgraphs relevant to the query.	Provide semantically coherent and explainable retrieval results by leveraging graph structure, enabling precise entity linking and relationship discovery.
Self-Reflective Validation and Fact Checking	
Agents perform internal evaluation of retrieved content by cross-referencing multiple sources, checking consistency, and assessing confidence scores. They may trigger additional searches or tool calls to resolve contradictions or verify claims.	Enhance result reliability and user trust by automatically validating facts, reducing hallucinations, and flagging uncertain information.

Table 1: Possible tasks for Agentic AI in Information Retrieval

Dataset	Description	Data Type	Task	Links*
The Pile	The Pile is an extensive, diverse dataset comprising roughly 800GB of text drawn from sources like ArXiv, GitHub, Wikipedia, and more. It has been meticulously compiled to offer a wide spectrum of writing styles and subject matter, making it ideal for training large-scale language models.	Textual	Train, Fine-tuning, NLP	Kaggle Paper Official Site
Common Crawl	Common Crawl aggregates billions of web pages scraped on a monthly basis, offering a true web-scale dataset. Its vast collection captures diverse content from across the internet, making it a foundational resource for training robust language models.	Textual	Search Engine, Web Language Models	Kaggle Paper Official Site
WikiText	WikiText is an open-source language modeling dataset derived from high-quality Wikipedia articles. It retains the rich structure and linguistic complexity found in editorial content, offering models a challenging environment to learn long-range dependencies. It also features a far larger vocabulary and retains the original case, punctuation and numbers.	Textual	Training, Benchmarking, Fine-tuning	Kaggle Paper Official Site
OpenWebText	OpenWebText is an open-source effort to recreate the WebText dataset originally used by OpenAI for language modeling. Compiled from web pages linked on Reddit, it provides a diverse collection of high-quality internet text.	Textual	Train, Fine-tuning, NLP	Kaggle GitHub Official Site
LAION-5B	LAION-5B is an enormous dataset containing 5.85 billion image-text pairs, providing an unprecedented resource for multimodal AI. Its scale and diversity support the training of cutting-edge text-to-image models like Stable Diffusion and DALL·E.	Image, Text	Training, Multimodal AI, Generative AI	Kaggle Paper Official Site

*Always check for eventual updates of links and datasets via a web search

Table 2: Commonly-used dataset for Agentic AI and Generative AI

evaluation.

Chapter 1: Introduction to Agentic AI in Information Retrieval

This chapter provides an overview of agentic artificial intelligence and its transformative potential for modern information retrieval systems. It introduces the core concepts of autonomy, goal-directed planning, tool integration, and adaptive reasoning, and discusses the challenges of retrieving relevant information in complex, large-scale corpora. The chapter frames the central problem—designing an IR agent that can interpret user intents, decompose multifaceted queries, and orchestrate multi-step retrieval workflows—and outlines the objectives of the proposed agentic IR framework.

Chapter 2: Data and Preprocessing

This section describes the datasets and knowledge sources relevant to agentic information retrieval. It reviews common IR corpora alongside structured resources (knowledge graphs, APIs) and details the preprocessing techniques required to support downstream agentic reasoning.

Chapter 3: Retrieval Methodologies

This chapter examines the spectrum of agentic retrieval approaches. For each methodology used, the underlying mechanisms are analyzed in terms of their advantages and suitability for different IR scenarios. The role of large language models, reinforcement learning, and external tool integration is highlighted throughout.

Chapter 4: System Implementation and Evaluation

This chapter details the design and implementation of the agentic IR system, describing its core modules—query planner, tool-invocation manager, retrieval engine, and validation component—and their interactions. It discusses architectural considerations, indexing strategies, and orchestration of external APIs for real-time data enrichment. The evaluation framework is defined, including benchmarks, relevance metrics, and agent-specific measures, and results are compared against baseline IR systems.

Chapter 5: Conclusions and Future Work

The final chapter summarizes the key contributions of integrating agentic AI into information retrieval, reflects on empirical findings, and discusses broader implications for search and knowledge management. It reviews the framework's limitations and outlines directions for future research, including multi-agent coordination, enhanced fact-checking modules, and application to domain-specific retrieval (e.g., legal, biomedical).

Possible Thesis Extention

The exam project can be extended into a **Master's thesis** with several advanced research directions. Some examples of possible Master's thesis directions are explained in detail below.

Agentic Query Decomposition for Spatio-Temporal Analysis

Approach : Designing an **agentic query planner** that autonomously decomposes complex, time-series Earth observation requests into a sequence of targeted retrieval subtasks (e.g., "identify deforestation hotspots, then retrieve seasonal change imagery").

Application : Users pose high-level, spatio-temporal questions (e.g., "Track urban growth and vegetative health in the Po Valley over the last decade"), and the agent orchestrates multi-step searches, combining CBIR, change detection, and metadata queries.

Method :

- Implement a **planning agent** based on an LLM with hierarchical task decomposition.
- Integrate **temporal retrieval modules** that call different archives/APIs for each time slice.

- Use a **tool-invocation manager** to coordinate CBIR, change detection, and multi-modal retrieval agents in sequence.

Tool-Augmented Agentic Retrieval with Geo-Analytics

Approach : Embedding an **agentic orchestration layer** that dynamically selects and invokes specialized geo-analytics tools (e.g., NDVI calculators, DEM processors, climate model APIs) within the retrieval pipeline [6].

Application : A researcher asks, "Show areas of significant vegetation stress correlated with elevation changes," and the agent fetches imagery, runs analytics, and synthesizes results.

Method :

- Develop a **tool registry** mapping LLM tool-calls to geo-processing functions (e.g., QGIS scripts, API endpoints).
- Implement **real-time API connectors** for on-the-fly computation (e.g., Google Earth Engine).
- Enable **chain-of-thought prompts** in the LLM to determine when to call each geo-tool and how to integrate outputs.

Self-Reflective Fact-Checking in Multi-Modal Retrieval

Approach : Incorporating a **self-validation agent** that cross-references retrieved satellite images and metadata against external authoritative sources (e.g., land-use registries, scientific publications) before presenting results.

Application : When asked, "Verify recent flood extent in the Danube Delta," the system retrieves imagery, then autonomously validates the observations against hydrological reports and machine-read publications.

Method :

- Implement multi-source **cross-verification routines** using both RAG and structured-data APIs.
- Use an LLM-based **confidence scoring module** that triggers additional searches for low-confidence findings.
- Present a **fact-check summary** that includes provenance annotations and uncertainty estimates.

Agentic AI Systems for Software Vulnerabilities[1]

Approach : Deploying an **autonomous vulnerability agent** that continuously audits software artifacts, reasons about potential threats, and proposes mitigations by simulating adversarial behaviors and consulting exploit databases.

Application : When tasked with "Identify exploitable logic flaws in a legacy authentication module," the system inspects the code, tests multiple execution paths, compares outputs with expected invariants, and consults CVE databases to validate its findings.

Method :

- Leverage **code-aware LLMs** to interpret software semantics and identify risky patterns across language boundaries.
- Employ **simulation-based probing** to test exploit chains, guided by reinforcement learning and symbolic execution.
- Integrate a **patch synthesis module** that drafts secure code replacements, referencing past vulnerability resolutions.

References

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